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REQUEST FOR COMMENTS — RFC-002

AI Mechanic

Forensic Runtime Assurance Layer **DRAFT**

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Author:

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RFC-001: CIEMS™ Defense AI Systems Assurance & Governance Layer

Approved Refinement:

Dar-z Morris, June 15, 2026

 INTERNAL USE NOTICE

AI Mechanic is an **internal module**. It is **NOT** presented to buyers as a standalone product. Externally, CIEMS + AI Mechanic = **ONE unified offer: "Defense AI Systems Assurance & Autonomy Validation Layer."** This document defines internal architecture, capability scope, and integration points with CIEMS only. Any external disclosure of the modular distinction between CIEMS and AI Mechanic requires explicit authorization.

Abstract

AI Mechanic is the forensic runtime assurance engine underlying the CIEMS governance architecture. Where CIEMS governs decision authority and traceability at the command layer, AI Mechanic operates at the **system execution layer** — continuously monitoring, auditing, and forensically analyzing AI runtime behavior to detect drift, failure, and integrity violations before they propagate into mission-critical decisions.

Together, **CIEMS (governance layer) + AI Mechanic (runtime layer)** = the complete Defense AI Systems Assurance stack. This RFC defines the internal scope, architecture, capability differentiators, and open design questions for AI Mechanic as a subsystem of the CIEMS architecture.

1. Problem Statement — Runtime Layer

Current defense AI deployments lack critical runtime-level assurance infrastructure. The gaps identified below represent operational risk at the execution layer that CIEMS governance alone cannot address without a dedicated forensic runtime engine:

Gap	Operational Risk
Runtime Behavioral Integrity	Systems run without continuous

Gap	Operational Risk
Monitoring	comparison against certified behavioral baselines, leaving drift undetected until failure occurs.
Forensic Auditability	When AI systems fail or produce anomalous outputs, there is no forensic record of execution state at the time of failure, preventing accountability and root-cause analysis.
Intent Drift Detection	Autonomous systems gradually diverge from mission-aligned behavior with no detection mechanism until the divergence produces a consequential output.
Failure Root-Cause Traceability	Post-incident analysis cannot reconstruct the system state that produced a failure, preventing corrective action and repeat-failure mitigation.
Workflow Integrity Verification	Multi-step AI workflows have no checksum or integrity verification across handoff points, creating exposure to corruption or injection between pipeline stages.

AI Mechanic is purpose-built to close each of these gaps at the execution layer, operating in continuous concert with the CIEMS governance stack.

2. What AI Mechanic Does — Internal Scope

AI Mechanic is the **runtime execution watchdog** for the CIEMS stack. It operates in four primary functional modes, each targeting a distinct assurance domain:

2.1 Continuous Runtime Monitoring

- Behavioral baseline capture on system initialization and re-certification events
- Continuous delta analysis against certified behavioral baseline across all runtime cycles
- Deviation thresholds configured per mission type, platform class, and risk level

- Real-time alert propagation to the CIEMS Governance Interface Dashboard (GID)

2.2 Forensic Audit Engine

- Immutable execution log capture at configurable granularity (per-cycle, per-decision, per-event)
- State snapshot archiving at all decision boundary events
- Chain-of-custody preservation for post-incident forensic reconstruction
- Timestamped audit trails compliant with DoD audit requirements and applicable federal standards

2.3 Intent Drift Detection

- Mission parameter alignment scoring evaluated on a continuous basis against defined objectives
- Semantic drift detection in natural language AI output streams
- Behavioral clustering to identify emergent off-mission patterns prior to consequential divergence
- Escalation protocols when drift exceeds mission-defined thresholds, routed to CIEMS governance layer

2.4 Workflow Integrity Verification

- Cryptographic integrity checksums applied at all multi-agent workflow handoff points
- Input/output consistency validation across sequential pipeline stages
- Tamper detection for AI workflow injection or manipulation between execution steps
- End-to-end traceability from input ingestion through to final decision output

3. Integration with CIEMS

AI Mechanic does not operate independently. It is a **subsystem of the CIEMS architecture**, with defined integration points across each CIEMS layer. The table below maps each CIEMS architectural layer to its corresponding AI Mechanic integration function:

CIEMS Layer	AI Mechanic Integration Point	Direction
Cognitive Decision Governance Layer (CDGL)	Feeds runtime anomaly data into the decision audit trail; flags decisions made under degraded or drifted system state	AI Mechanic → CDGL
Runtime Assurance Module (RAM)	AI Mechanic <i>is</i> the RAM subsystem — primary execution engine for all runtime verification functions	Native subsystem
Systems Clarity Engine (SCE)	Supplies workflow integrity verification data to clarity mapping; provides execution-layer context for explainability outputs	AI Mechanic → SCE
Economic Memory Layer (EML)	Forensic logs feed institutional failure memory; enables pattern recognition across historical anomalies and drift events	AI Mechanic → EML
Governance Interface Dashboard (GID)	AI Mechanic surfaces all runtime status, active alerts, drift scores, and integrity flags to the command layer in real time	AI Mechanic → GID

Integration Principle

Every AI Mechanic output is consumed by a CIEMS layer. AI Mechanic generates no standalone external-facing interface. All visibility surfaces through CIEMS governance controls.

4. Deployment Architecture

4.1 Deployment Modes

Mode	Description	Primary Use Case
Embedded Mode	AI Mechanic runs as a co-process alongside the monitored AI system at the platform or host level	Autonomous platforms, edge-deployed tactical AI
Network Mode	Monitoring via API interception and log streaming; does not require co-location with monitored system	Enterprise AI systems, data center-hosted models, connected analysis platforms
Forensic-Only Mode	Post-hoc analysis of existing AI execution logs without live monitoring capability	Incident investigation, compliance review, historical audit of legacy AI deployments

4.2 Supported System Types

- Autonomous vehicle and platform AI (air, ground, and maritime domains)
- Intelligence analysis and multi-source fusion AI systems
- Natural language decision support and command advisory systems
- Logistics optimization and defense supply chain AI
- Cyber threat detection, attribution, and response AI

4.3 Classification Compatibility

- Designed for deployment across **UNCLASSIFIED, CUI, and SECRET** network environments
- Air-gap compatible for classified and sensitive compartmented network environments
- **FIPS 140-2/3** compliant cryptographic modules for all integrity verification functions

5. Capability Differentiators vs. Existing Tools

The table below compares AI Mechanic against the current state of standard AI monitoring and observability tools available in both commercial and defense markets:

Capability	Standard AI Monitoring Tools	AI Mechanic
Forensic Auditability	Log-based, partial — gaps at decision boundaries	Full state-chain forensic reconstruction with chain-of-custody preservation
Intent Drift Detection	Threshold-based alerts only; no semantic or clustering analysis	Semantic drift detection + behavioral clustering for early-pattern identification
Workflow Integrity	None — no verification across pipeline handoffs	Cryptographic checksums at all multi-agent workflow handoff points
CIEMS Integration	N/A — not designed for governance-layer interoperability	Native — purpose-built as the RAM subsystem within CIEMS architecture
Classification Compatibility	Commercial only; not cleared for classified environments	Multi-classification, FIPS 140-2/3 compliant, air-gap capable
Root-Cause Reconstruction	Requires manual log correlation; incomplete state data	Automated state-snapshot replay; full pre-failure state reconstruction

6. RFC Open Questions — Internal

The following questions are open for internal review and resolution prior to Blueprint development. Respondents should direct comments to the RFC author through approved internal channels.

Q1. What granularity of runtime state capture is operationally viable at mission execution speed, and what are the latency constraints for different platform classes?

Q2. Which behavioral drift thresholds are appropriate for each autonomy level (Levels 1–5), and should thresholds be static or dynamically calibrated per mission context?

Q3. How should forensic logs be handled across classification boundaries — particularly when a monitored system transitions between network classification domains during operation?

Q4. What are the storage and compute overhead constraints for AI Mechanic in field-deployed, bandwidth-limited, or SWaP-constrained environments?

Q5. Should AI Mechanic be disclosed as a named component in external proposals and technical volumes, or remain fully internal with all external reference collapsed into the unified CIEMS offer?

7. Build Priority

Per approved execution order, the following sequencing governs AI Mechanic development artifacts:

#	Deliverable	Owner	Status
1	Visual System Map (external/internal architecture split)	Daniel	In Queue
2	RFC-001: CIEMS™ Defense AI Systems Assurance & Governance Layer	Jon	In Progress
2	RFC-002: AI Mechanic — Forensic Runtime Assurance Layer (this document)	Jon	In Progress
3	Blueprint: CIEMS Architecture Specification	Jon	Pending RFC-001 completion
3	Blueprint: AI Mechanic Forensic Runtime Specification	Jon	Pending RFC-002 completion
3	Blueprint:	Jon	Pending RFC

#	Deliverable	Owner	Status
	Deployment Model		completion

8. Execution Lock

EXECUTION LOCK — Approved Refinement: Dar-z Morris, June 15, 2026

CIEMS and AI Mechanic do NOT operate as parallel external products. Externally, they collapse into ONE unified offer: "Defense AI Systems Assurance & Autonomy Validation Layer." Internally they remain architecturally modular. This is non-negotiable per approved Dar-z Morris refinement, June 15, 2026. No external materials, proposals, capability briefs, or contract vehicles shall present AI Mechanic as a distinct product offering.

Appendix A — Glossary

Term	Definition
AI Mechanic	Internal forensic runtime assurance engine; designated as the Runtime Assurance Module (RAM) subsystem within the CIEMS governance stack. Not an external product.
CIEMS	Cognitive Intelligence & Enterprise Management System — the overarching governance architecture of which AI Mechanic is a subsystem. Externally presented as the unified Defense AI Systems Assurance & Autonomy Validation Layer.
Runtime Assurance	Continuous verification that an AI system's

Term	Definition
	live execution behavior conforms to certified behavioral baselines and defined mission parameters across all operating cycles.
Forensic Audit	Post-hoc or concurrent reconstruction of AI system execution state for accountability, root-cause analysis, and legal or regulatory chain-of-custody purposes.
Intent Drift	Gradual divergence of an AI system's runtime behavior from defined mission objectives, detectable through behavioral clustering analysis and semantic output monitoring.
Workflow Integrity	Verification that multi-step AI pipelines have not been tampered with, corrupted, or injected between sequential execution stages, enforced via cryptographic checksums at handoff points.
Behavioral Baseline	A certified reference representation of expected AI system behavior, captured at initialization or re-certification, against which continuous runtime delta analysis is performed.
Chain of Custody	Documented, tamper-evident record of AI execution state transitions, preserving the integrity and admissibility of forensic audit data for post-incident reconstruction.

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